

CURRICULUM VITAE

PERSONAL INFORMATION

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WORK EXPERIENCE

- Dates (Feb. 2026 - Present)
- Employer's name and locality **Micro1. Remote (USA).**
- Position Held *Geospatial data scientist & cartographer*
- Main activities and responsibilities

Geospatial specialist role focused on the design and evaluation of algorithms, data structures and quality for GIS and mapping platforms. Working at the intersection of cartography, data quality and AI development, contributing both technical expertise and domain knowledge to ensure the accuracy and reliability of geospatial products.

 - Inspect and validate final map compositions for accuracy, completeness and compliance with data integrity standards.
 - Collaborate with AI development teams as a geospatial subject matter expert, providing structured feedback to improve the performance and reliability of geospatial AI solutions.
- Dates (Nov. 2022 - May 2025)
- Employer's name and locality **Astara Innovation. Av. de Bruselas, 32, Alcobendas, Spain.**
- Position Held *Lead Data Scientist*
- Main activities and responsibilities

Data science role embedded in an urban mobility and connected vehicle environment. Conflated applied research, geospatial analysis and machine learning to extract actionable insights from large-scale location-based and telematics datasets.

 - Conducted applied research on urban mobility patterns using large-scale car floating data, applying geospatial tools (PostGIS, GeoPandas, Shapely) to model and optimise user driving behaviour.
 - Implemented and deployed geostatistical models — including universal kriging and spatial autocorrelation analysis — to evaluate and map air pollution levels across urban areas.
 - Developed dasymetric mapping workflows integrating Cell Data Records origin-destination data with OpenStreetMap via the Overpass API to produce high-resolution spatial distributions.
 - Designed an eco-driving score system from telematics data, enabling behavioral profiling of connected vehicle users to support sustainability-focused product features.
 - Close collaboration with product and development teams to design, develop, deploy and maintain mobility based products (APIs, Apps).

• Dates	(April 2022- Nov. 2022)
• Employer's name and locality	Centro de Investigación del Transporte (TRANSyT). No. 3 Prof. Aranguren st, 28040. Madrid, Spain.
• Position Held	<i>Postdoctoral researcher</i>
• Main activities and responsibilities	Postdoctoral research role within a university transport research centre, focused on the economics and policy of road pricing. Combines rigorous academic methodology with computational spatial analysis, producing reproducible, evidence-based outputs to inform both scientific debate and public policy. • Conducted scientific research and Performed a PRISMA-guided literature review on road pricing strategies. • Created reproducible Jupyter Notebooks performing Spatial Exploratory Data Analysis on diverse datasets.
• Dates	(Mar. 2020 - Present)
• Employer's name and locality	Instituto Geográfico Universitario. Remote (Dom. Rep.).
• Position Held	<i>Applied Researcher</i>
• Main activities and responsibilities	Conduct applied geographic and spatial research projects for private companies, public institutions and research organisations. • Earth observation analyses using satellite imagery and remote sensing techniques for territorial monitoring. • Produce land cover maps and technical reports to support decision making and project development. • Process, analyse and interpret geospatial datasets using GIS and remote sensing tools. • Collaborate with multidisciplinary teams to design research methodologies and deliver client-oriented solutions. • Develop web mapping applications and interactive geospatial visualisation platforms for data dissemination and decision support.
• Dates	(Mar. 2020 - Present)
• Employer's name and locality	Universidad Autónoma de Santo Domingo. Alma Mater Ave. Santo Domingo, Distrito Nacional (remote).
• Position Held	<i>Cartography and image processing lecturer</i>
• Main activities and responsibilities	University lecturer focused on cartography, GIS and remote sensing. Combines theoretical instruction with practical, software-based training to develop students' spatial analysis and image processing competencies. Works with undergraduate students, bridging foundational cartographic concepts with current industry tools and workflows. • Deliver hands-on training in industry-standard GIS software including QGIS, ArcGIS Pro and ArcMap, covering spatial analysis techniques and cartographic production workflows. • Guide students through satellite image pre-processing pipelines, including radiometric and atmospheric corrections, filtering, and band combination techniques. • Instruct supervised and unsupervised image classification methods, enabling students to extract meaningful land cover and thematic information from remote sensing data.

- Dates (May. 2016- Feb. 2022)
 - Employer's name and locality **E.T.S. I. Caminos Canales y Puertos.** Dpto. de Ingeniería del Transporte, Territorio y Urbanismo (UPM). No. 3 Prof. Aranguren st., 28040. Madrid, Spain.
 - Position Held *Road Safety researcher*
 - Main activities and responsibilities
 - Planned the collection and processed LiDAR data from Topcon IPS3 system, Leica C10, Leica Pegasus Backpack and PNOA LiDAR.
 - Owned the pipeline to exploit geospatial and non-geospatial data to conduct visibility evaluations.
 - Wrote research papers, reports, reviews, and summaries regarding the obtention of available sight distances on existing roads and its relationship with road safety.
 - Used LiDAR data and GIS tools to create realistic representations of the road environment and carry out virtual visibility assessments.
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- Dates (May. 2015 – July. 2015)
 - Employer's name and locality **Instituto Geográfico Nacional de España.** No. 3 Gral. Ibáñez de Ibero, 28003. Madrid, Spain.
 - Position Held *Earth observation intern*
 - Main activities and responsibilities
 - Documented satellite imagery selection and quality checks processes.
 - Created workflows for the processing and handling of large geospatial datasets.
 - Processed geospatial data with GIS software (FME, ArcGIS, QGIS).
 - Obtained territorial statistics of land cover classes from the data of the Land Use Information System (SIOSE).
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- Dates (Dec. 2014)
 - Employer's name and locality **Buque Oceanográfico Miguel Oliver.** Ministerio de Agricultura, Alimentación y Medio Ambiente, Spain.
 - Position Held *Scientific support*
 - Main activities and responsibilities
 - Processed data to create maps of Spain's Exclusive Economic Zone.
 - Characterized soil samples acquired with the trawl Baca 44/60 (with sediment sampling cylinder).
 - Verified observations from the following equipment: Konsberg Simrad EM302 multibeam echosounder and TOPAS PS018 parametric echo sounder.
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- Dates (Jun. 2006 - Jun. 2009)
 - Employer's name and locality **HISARQ (Hidráulicas, Sanitarias y Arquitectura).** N. 8, Teodoro Chaserau st., Dominican Republic.
 - Position Held *Junior engineer*
 - Main activities and responsibilities
 - Elaborated Environmental Impact Statements (EIS) and Assessments (EIA).
 - Contributed to project's technical documentation.
 - Used GIS software to evaluate environmental effects of civil engineering projects.

EDUCATION & TRAINING

Qualification Awarded	Civil engineering Ph.D.
Dates	(2016-2021)
Organization	Escuela Técnica Superior de Ingenieros de Caminos, Canales y Puertos. Universidad Politécnica de Madrid. <i>Madrid, Spain.</i>
Program	Doctorado en Sistemas de Ingeniería Civil (DOSIC) Visiting scholar at Iowa State University (Jul-Sept. 2021)
Qualification Awarded	Data Analyst NanoDegree
Dates	(2020)
Organization	Udacity
• Qualification Awarded	Master in Geodesy Engineering and Cartography
Dates	(2013-2015)
Organization	Escuela Técnica Superior de Ingenieros en Topografía, Geodesia y Cartografía. Universidad Politécnica de Madrid. <i>Madrid, Spain.</i>
Principal subjects	Geospatial information acquisition, processing, and diffusion techniques.
• Qualification Awarded	Civil Engineer
Dates	(2004– 2012)
Organization	Universidad Autónoma de Santo Domingo (UASD). <i>Sto. Domingo, Dom. Rep.</i>
Principal subjects	Structural analysis, highway's geometric design, environmental studies.

LANGUAGES

Spanish: Native | **English** level: C1 | **Italian:** A1

SKILLS & STACK

Data analysis *Python (Pandas, Numpy, GeoPandas, Shapely, H3, toblor)*

Data extraction and cleaning *(PostgreSQL, PostGIS)*

Data visualization *(Matplotlib, Seaborn, Leaflet)*

Version control *(GitHub)*

Containerization *(Docker)*

GIS software: ArcGIS Pro, ArcMap, QGIS, Global Mapper, FME, Las Tools, MDTOPX.

Cloud infrastructure and deployment *(AWS)*

Product development tracking *(Linear)*

RELEVANT PUBLICATIONS

González-Gómez, K.; Iglesias, L.; Rodríguez-Solano, R.; Castro, M. (2019): "Framework for 3D Point Cloud Modelling Aimed at Road Sight Distance Estimations". *Remote Sens*, 11, 2730.

González-Gómez, K., Castro, M. (2018): Suitability testing of LiDAR processing Software aimed at 3-D sight distance estimations. *Transportation Research Procedia*, 33, 163–170.
